

Coverage Path Planning

Impact of Drone Endurance on Multi-Operation Planning

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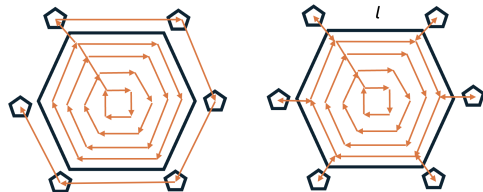
OR Conference · 15–17 min talk + Q&A

Joint work: Amorosi – Dell'Olmo – Puerto – Valverde · Gurobi 11.0 + Python · reproducible by seed

Introduction: the problem

Drone coverage path planning (CPP):

- One drone must cover a set of **disjoint polygonal regions** in the plane
- **Limited battery**: each flight (operation) starts and ends at the depot, where the battery is swapped
- Each region can be covered along **spiral chains at different altitudes**: higher altitude = fewer passes, but less air density and more wind
- **Preemption allowed**: the drone may leave a region unfinished and resume it later
- **New implementation**: entry/exit points placed *anywhere* along the rings, not fixed



Existing approaches

Our approach

Left: preemption allowed. Right: two-stage (each region in one go).
Triangle + 3 satellites: gap = $l/6$.

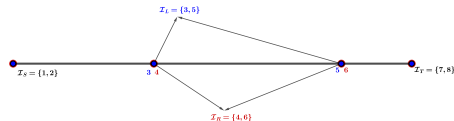
Applications: aerial inspection, precision agriculture, 3D mapping, disaster response.

Objective

Minimise **total distance/time** to cover everything, respecting endurance $W^{\max}$ per operation.

The general problem

- $\mathcal{R} = \{1, \dots, R\}$ disjoint polygonal regions; depot $v_0 = (0, 0, 0)$; endurance $W^{\max}$
- Each region r : set $\mathcal{T}_r$ of chains, chain t at altitude h_t with segments $\{[Q^s, Q^{s+1}]\}_{s \in \mathcal{S}_t}$
- **Decision**: one chain per region (α_t), entry/exit points on it, order of coverage
- **Preemption**: the drone may interrupt a region and resume it later (up to $R - 1$ interruptions)
- Operations $\mathcal{O}$: tours starting/ending at the depot



Candidate points along a chain: start $\mathcal{I}_S$, launch $\mathcal{I}_L$, retrieve $\mathcal{I}_R$, end $\mathcal{I}_T$

Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$: intra-chain edges $\mathcal{E}^{int}$, inter-region arcs $\mathcal{E}^{ext}$.

Feasibility

Every region fits alone in one operation (depot–region–depot $\leq W^{\max}$).

Decision variables

As in the paper, the model has **four families of constraints** — (i) location, (ii) drone path, (iii) distance, (iv) consumption — plus the objective (total distance). The main decision variables are:

Binary and integer:

- μ_v^{ts} : vertex v lies on segment s of chain t (location)
- α_t : chain t is the one selected for its region
- $x_{vv'}^o$: the drone traverses (v, v') in operation o (routing)
- y_v^o : vertex v is visited during operation o
- ζ^o : everything is covered before operation o starts
- k^o : number of vertices (or edges) covered in o
- $\varphi_{rr'}, \psi_{rr'}$: inter-region flight altitude and vertical ordering (energy)

Continuous:

- P_v : coordinates of vertex v on the selected chain (decision, not input!)
- $\gamma_v \in [0, 1]$: position of v inside its segment; λ_v : global chain parametrisation
- d_{fs_v}, e_{fs_v} : cumulative distance / energy from the chain start
- $d_{vv'}, d_{vv'}^{xy}, d_{vv'}^z$: distance and its horizontal/vertical parts
- $w_{vv'}$: energy consumed on (v, v')
- ℓ_{uv}^o : linearisation of $x_{uv}^o \cdot d_{uv}$ (objective) and $x_{uv}^o \cdot w_{uv}$ (endurance)
- u_v^o : position of v in the tour of o (MTZ subtour elimination)

Formulation overview — four families

Four constraint families + objective (total distance)

(i) Location

$P_v, \lambda_v, \mu, \gamma, \eta, \alpha$

places vertices on segments

(ii) Routing

$x_{vv'}^o, y_v^o, \zeta_v^o, k_v^o, u_v^o$

operations, no subtours

(iii) Distances

$d_{uv}, d^{xy}, d^z, dfs_v + \text{SOC}$

Euclidean + vertical

(iv) Energy

$w_{uv}, efs_v, \varphi, \psi, \ell_{uv}^o$

drag + endurance

$$\min W = \sum_{(u,v)} \sum_o \ell_{uv}^o \quad \text{s.t. Loc-C, } D\text{-C, Dist-C, } w\text{-C, Endurance}$$

ℓ_{uv}^o linearises $x_{uv}^o d_{uv}$ (objective) and $x_{uv}^o w_{uv}$ (endurance) via big- M .

(i) Location constraints

Vertex $v = (r, i)$ on segment s of chosen chain t , offset $\gamma_v \in [0, 1]$:

$$P_v = \sum_{t,s} [\mu_v^{ts} Q^s + \eta_v^{ts} (Q^{s+1} - Q^s)] \quad (\text{Loc-C}_1)$$

$$P_v^z = \sum_{t,s} \mu_v^{ts} h_t \quad (\text{Loc-C}_1^z)$$

$$\sum_s \mu_v^{ts} = \alpha_t, \quad \sum_t \alpha_t = 1 \quad (\text{Loc-C}_{2,3})$$

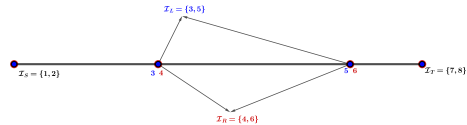
$\eta_v^{ts} = \gamma_v \mu_v^{ts}$ via Fortet.

Parametrisation $\lambda_v \in [0, S_t]$:

$$\lambda_v = \sum_{t,s} (s-1) \mu_v^{ts} + \gamma_v \quad (\text{Loc-C}_4)$$

$$\lambda_v = 0 \quad (v=(r, 1), (r, 2)) \quad (\text{Loc-C}_5)$$

$$\lambda_v = \sum_t S_t \alpha_t \quad (v=(r, 2R+1), (r, 2R+2)) \quad (\text{Loc-C}_8)$$



$$\mu_v^{t0} = \alpha_t, \quad \mu_v^{t, S_t-1} = \alpha_t \quad (\text{Loc-C}_{5b, 8b})$$

$$\lambda_u \leq \lambda_v \quad (u, v) \in \mathcal{E}_{RL}^{int} \quad (\text{Loc-C}_6)$$

$$\lambda_u = \lambda_v \quad (u, v) \in \mathcal{E}_{LR}^{int} \quad (\text{Loc-C}_7)$$

Note

RL = forward, LR = zero-length return ($\lambda_u = \lambda_v$).

(ii) Routing constraints

$x_{vv'}^o$: traverse $v \rightarrow v'$ in o ; y_v^o : visit v in o ; ζ^o : all done.

$$\sum_{v'} x_{v_0 v'}^o = 1 - \zeta^o \quad (D-C_1)$$

$$\sum_{v'} x_{v' v_0}^o = 1 - \zeta^o \quad (D-C_3)$$

$$\sum_{v'} x_{v' v}^o - \sum_{v'} x_{v v'}^o = 0 \quad (D-C_2)$$

$$\sum_{v'} x_{v v'}^o = y_v^o, \quad \sum_{v'} x_{v' v}^o = y_v^o \quad (D-C_{4,5})$$

$$\sum_o y_v^o = 1 \quad (D-C_6)$$

$$\sum_o x_{v v'}^o = 1 \quad (v, v') \in \mathcal{E}_{RL}^{int} \quad (D-C_8)$$

$$\text{MTZ: } u_v^o - u_{v'}^o + 1 \leq |\mathcal{V}'|(1 - x_{v v'}^o) \quad (D-C_7)$$

Valid inequalities

$$\zeta^o \leq \zeta^{o+1} \quad (\text{Mon.})$$

$$k^o = \sum_v y_v^o \quad (k-C)$$

$$\sum_{o' < o} k^{o'} \geq |\mathcal{V}'| \zeta^o \quad (\text{VI-C}_1)$$

$$k^o \geq 1 - \zeta^o \quad (\text{VI-C}_2)$$

Meaning

$\zeta^o=1 \Rightarrow$ idle operation; monotonicity removes symmetry.

(iii) Distance constraints

Intra-region (DFS trick): avoids $\beta_{vv'}^{ss'} = \mu_v^s \mu_{v'}^{s'}$

$$dfs_v = \sum_{t,s} [\text{cum}(s) \mu_v^{ts} + L(s) \eta_v^{ts}] \quad (\text{DFS})$$

$$d_{uv} = dfs_v - dfs_u \quad (u, v) \in \mathcal{E}_{RL}^{int} \quad (\text{Dist-C}_1)$$

$$d_{uv} = 0 \quad (u, v) \in \mathcal{E}_{LR}^{int} \quad (\text{Dist-C}_2)$$

dfs_v = cumulative distance from chain start.

Depot legs:

$$d_{v_0v} = d_v^{xy} + d_v^z, \quad d_v^{xy} \geq \|(P_v^x, P_v^y)\|_2, \quad d_v^z \geq |P_v^z| \quad (\text{Dep-C})$$

Inter-region: $xy + z$

$$d_{vv'} = d_{vv'}^{xy} + d_{vv'}^z \quad (\text{Dist-C}_3)$$

$$d_{vv'}^{xy} \geq \|(P_v^x, P_v^y) - (P_{v'}^x, P_{v'}^y)\|_2 \quad (\text{Dist-C}_4 \text{ SOC})$$

$$d_{vv'}^z \geq |P_v^z - P_{v'}^z| \quad (\text{Dist-C}_{5,6})$$

Key

Dist-C₄ is a **second-order cone**. Both bounds tight at optimum. DFS/EFS replaces $O(|\mathcal{V}'|^2 |\mathcal{S}_t|^2)$ bilinear β variables.

(iv) Energy — intra-region

EFS mirrors DFS:

$$efs_v = \sum_{t,s} C_{ts} [\text{cum}(s) \mu_v^{ts} + L(s) \eta_v^{ts}] \quad (\text{EFS})$$

$$w_{uv} = efs_v - efs_u \quad (u, v) \in \mathcal{E}_{RL}^{int} \quad (w\text{-C}_1)$$

$$w_{uv} = 0 \quad (u, v) \in \mathcal{E}_{LR}^{int} \quad (w\text{-C}_2)$$

$C_{ts} = E^{xy} \rho_t \nu_{ts}^d$ precomputed energy per metre.

Depot legs:

$$\begin{aligned} w_{v_0 v} &= E^{xy} \rho_0 \nu^{xy} d^{xy} + \frac{1}{2} E^z \rho_0 \nu^z d^z \\ w_{vv_0} &= E^{xy} \rho_0 \nu^{xy} d^{xy} \quad (\text{no recuperation}) \end{aligned}$$

Saving

Same trick as DFS: exact elimination of bilinear $\mu_v^s \mu_v^{s'}$; no approximation.

efs_v = cumulative energy from chain start. Ascent pays E^z , descent does not.

(iv) Energy — inter-region

With $\varphi_{rr'}, \psi_{rr'} \in \{0, 1\}$ (flight altitude & vertical ordering):

$$w_{vv'} = \varphi_{rr'} \left(E^{xy} d_{vv'}^{xy} \sum_t \alpha_t \rho_t |\nu_{rr't}^d| + E^z \psi_{rr'} d_{vv'}^z \frac{1}{2} \sum_{\tau} \sum_t \alpha_t \rho_t \nu^z \right) \\ + (1 - \varphi_{rr'}) \left(E^{xy} d_{vv'}^{xy} \sum_t \alpha_t \rho_t |\nu_{rr't}^d| + E^z (1 - \psi_{rr'}) d_{vv'}^z \frac{1}{2} \sum_{\tau} \sum_t \alpha_t \rho_t \nu^z \right) \quad w\text{-C}_3$$

All products $\varphi\psi\alpha d$ linearised via Fortet (`_lin_bb`, `_lin_bc`), 4 constraints each.

Complexity

Most technical part; fully linearised.

Endurance & objective

Endurance (big- M):

$$\ell_{uv}^o \leq x_{uv}^o M$$

$$\ell_{uv}^o \leq w_{uv}$$

$$\ell_{uv}^o \geq w_{uv} - (1 - x_{uv}^o)M$$

$$\ell_{uv}^o \geq 0$$

$$\sum_{u,v} \ell_{uv}^o \leq W^{\max} \quad \forall o$$

ℓ_{uv}^o linearises $x_{uv}^o w_{uv}$.

(End-C₁)

(End-C₂)

(End-C₃)

(End-C₄)

(Endurance)

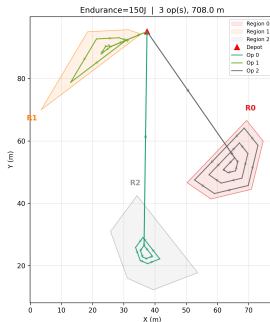
Objective

$\min W = \sum_{u,v} \sum_o \ell_{uv}^o$, ℓ_{uv}^o now linearises $x_{uv}^o d_{uv}$
(same variable, both bounds).

Resulting model: **MIQP** (SOC + fully linearised bilinears).

Problem setup

- 3 polygonal regions, offsets generated by uniform scaling
- Selected chains at heights: $R0 = 15$ m, $R1 = 40$ m, $R2 = 65$ m
- Drone: front area 0.1 m^2 , drag coef 0.3, cruise speed 15 m/s , vertical speed 5 m/s
- Pseudo-energy model: $E = 0.5 \cdot C_d \cdot A \cdot \rho \cdot v \cdot d$ (Joules)
- One drone, one depot: after each operation it returns to swap the battery; endurance $W^{\max}$ caps every operation
- **New implementation:** the drone may enter or leave each ring *at any point* of its perimeter

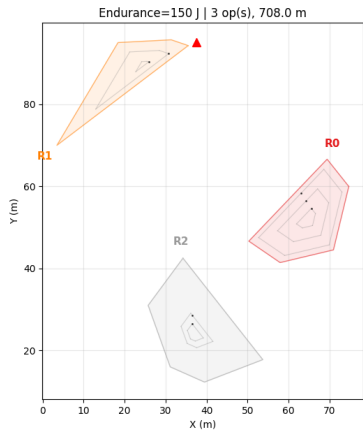


The following snapshots show the drone's path on the 3-region instance:

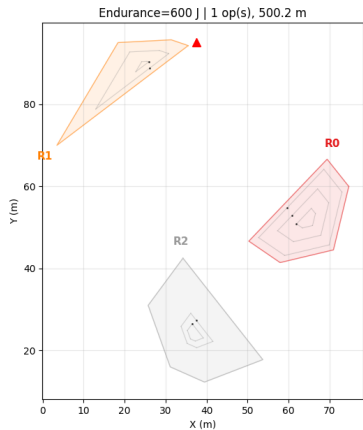
- **Solid lines:** RL (coverage) edges along the chain
- **Dotted lines:** transitions between segments or regions
- **Colors:** each operation in a different color
- All operations start from the depot ($\triangle$) and return to it
- **Markers:** entry/exit points, placed freely along the rings

Question: how many operations does a given endurance $W^{\max}$ require, and what total distance does the drone travel?

Endurance = 150 J — 3 operations



Endurance = 600 J — 1 operation



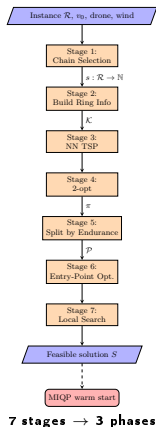
Showcase summary

Endurance (J)	Operations	Distance (m)	Gap (%)
150	3	708.0	1.8
200	3	706.0	1.8
250	2	611.5	1.9
300	2	596.6	24.5
400	1	500.5	2.0
600	1	500.2	2.0

More endurance $\Rightarrow$ fewer depot returns $\Rightarrow$ shorter total tour.

New implementation with free entry/exit points on the rings; Gurobi TL 45 s, MIPGap 2% (300 J run ended at the time limit).

Multi-phase heuristic: overview



Three phases:

- 1 **Pre-processing**: chain selection + ring geometry
- 2 **Construction**: giant tour + greedy endurance split
- 3 **Improvement**: coordinate descent + 2-opt & Or-opt

Goal

Feasible solution in < 0.6 s as **warm-start** for the MIQP (10–100 $\times$ faster first incumbent).

Heuristic: phases 1–3 in detail

Phase 1 – chain selection:

- Keep only max-resolution chains
- Proxy $c(t) = \sum \text{perim}(\kappa) + 0.5 (d_{xy} + h_t)$ per region

Phase 2 – construction:

- Nearest-neighbour on 3D centroids from v_0 ; expand rings interior→exterior
- Greedy split: add rings while energy $\leq W^{\max}$, else close operation

Phase 3 – improvement:

- Entry-point coordinate descent (60 samples, 8 passes)
- **2-opt** intra-operation; **Or-opt** inter-operation (relocate ring if both ops feasible and energy drops)
- Final solution feeds Gurobi Start (`set_warm_start`)

$\gamma = 0.5$, $N_{\text{samples}} = 60$, $N_{\text{coord}} = 8$, $N_{\text{passes}}^{\text{op}} = 10$.

Experimental design

Testbed:

- $n_r \in \{1, 2, 3, 5, 8, 10\} \times 3$ height levels $\times 5$ seeds $\times 6$ endurance values = **540 configs**
- Methods: heuristic (H), Vertex-Approach MIQP cold (R), warm-started ($R+WS$)
- Heights: L1 {30}, L2 {20,50}, L3 {15,35,60} m

Status (31 Aug 2026): Vertex-Approach sweep complete — 540 configs $\times$ 3 methods = 1620 runs (1703 CSV rows).

Endurance calibration: cheapest single-region tour;

rounded to 300 J (5% margin).

Factor	Raw	Rounded
$E_{\min}$	245.9	—
1.0	300	300
1.5	450	450
2.0	600	600
3.0	900	900
5.0	1500	1500
8.0	2400	2400

Solver: Gurobi 11.0, MIPGap=0, TL=1800 s, 1 thread.

All means below computed from
`compare_results/results.csv`.

Results — aggregated means (vertex, complete)

Table: Aggregated results for the Vertex-Approach (MTZ) formulation. Mean objective and solve time averaged over height levels ($n_h = 1, 2, 3$), endurance levels ($E = 300\text{--}2400$ J) and 5 seeds. MIP gap at the 1800 s limit, counting runs with no solution as 100% gap. H = heuristic standalone.

n_r	H		R (Vertex-Approach)			$R+WS$ (Vertex-Approach+WS)		
	Obj.	Time (s)	Obj.	Time (s)	Gap (%)	Obj.	Time (s)	Gap (%)
1	216.0	0.03	210.1	111.5	1.3	210.4	112.9	2.4
2	375.0	0.06	364.3	454.3	4.2	363.8	436.6	4.0
3	517.0	0.10	485.9	1088.8	8.0	485.8	1094.0	8.3
5	989.2	0.21	992.4	1790.3	26.8	943.4	1800.1	24.8
8	1676.2	0.41	2163.7	1800.2	55.5	1647.5	1800.2	39.2
10	2175.2	0.54	3283.6	1800.3	69.5	2017.2	1800.3	46.7

Means over n_h , E , seeds (540 configs). H = heuristic; R = vertex MIQP; $R+WS$ = warm-started.

Read: gap 0.1% at $n_r = 1$ (optimality proven) $\rightarrow$ 61% at $n_r = 10$ cold, 42% warm. Heuristic 0.03–0.54 s always feasible.

Results — endurance effect ($n_r = 3$, means)

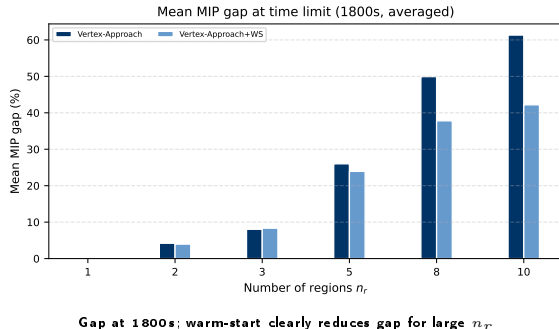
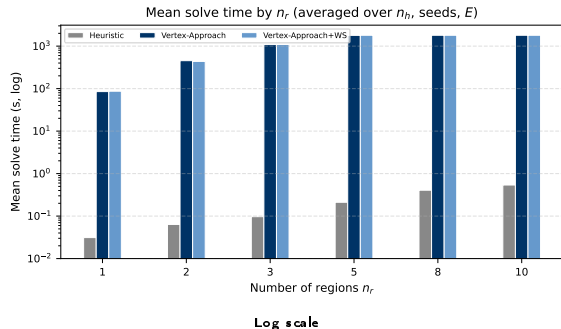
Table: Effect of endurance E for $n_r = 3$ (averaged over $n_h = 1, 2, 3$ and 5 seeds). Tight E implies more operations and higher total distance.

E (J)	H		R			$R+WS$		
	Obj.	Time	Obj.	Gap	Time	Obj.	Gap	Time
300	570.8	0.10	524.7	9.5	1122.7	525.8	9.5	1066.8
450	528.3	0.10	492.2	9.7	1121.8	492.2	10.8	1120.2
600	500.7	0.09	474.3	6.5	1062.7	474.3	7.6	1143.7
900	500.7	0.09	474.5	7.5	1070.7	474.3	7.4	1110.3
1500	500.7	0.10	475.2	7.0	1037.1	474.3	7.4	1035.5
2400	500.7	0.10	474.5	8.1	1117.7	474.3	7.1	1087.4

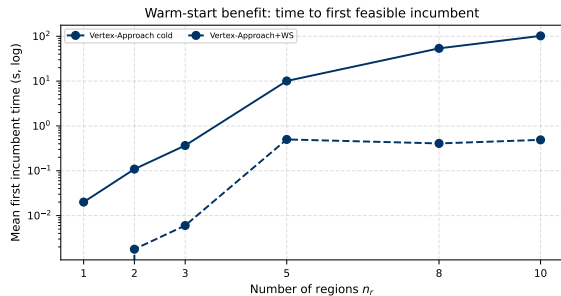
$n_r = 3$ fixed, averaged over n_h and seeds.

Tight $E = 300 \Rightarrow \approx 525$ m; $E \geq 600 \Rightarrow \approx 474$ m (endurance-unconstrained plateau). Same conclusion as the 60–120 J showcase, now on 90 instances.

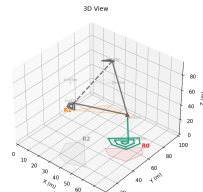
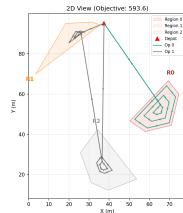
Comparative figures — time & gap



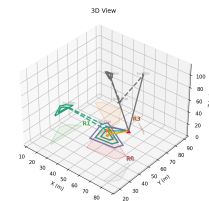
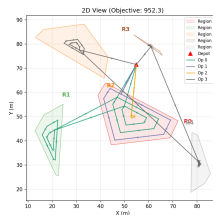
Warm-start benefit & visual solutions



Time to first feasible incumbent, $n_r = 1..10$ (mean over seeds/ n_h/E); dashed = warm-start



2 regions, 2 altitudes; colour = operation



5 regions; heuristic avoids crossings, MIQP refines

Warm-start saves 10–100× to the first feasible incumbent, most for tight endurance.

Vertex-Approach analysis — the complete case

540 configs $\times$ 3 methods = 1620 runs, all completed (31 Aug 2026).

- **Optimality:** gap $\approx 0.1\%$ for $n_r=1$ (proven); $n_r \leq 3$ within $\approx 8\%$ at the 1800 s limit
- **Heuristic:** 0.03–0.54 s, *always* feasible, within 2–6% of the exact objective on provable instances
- **Warm-start:** 10–100 $\times$ faster to the first feasible incumbent; final gap at $n_r=10$ drops 61% $\rightarrow$ 42%
- **Endurance effect** ($n_r=3$): $E=300$ J $\Rightarrow \approx 525$ m vs $E \geq 600$ J $\Rightarrow \approx 474$ m; plateau $\Rightarrow$ endurance-unconstrained
- **Scalability:** solve time grows with n_r ; all $n_r \geq 5$ hit the time limit, WS recovers ≈ 10 –20% gap

All means from `compare_results/results.csv` (`table_results_summary`, `table_results_by_endurance`, `compare*_vertex`).

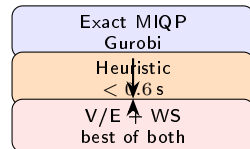
Conclusions & future work

Contributions:

- 1 MIQP with **preemption + altitude + endurance** in one model (SOC + big- M), one family per slide
- 2 Equivalent **Edges-Approach** DFJ reformulation
- 3 Multi-phase heuristic < 0.6 s that **warm-starts** the exact solver
- 4 Mean-based results on the full 540-config Vertex-Approach sweep

Future:

- Multi-drone heterogeneous fleet; dynamic wind
- VNS/LNS metaheuristics; real missions



```
python compare_methods.py -resume
```

Data: compare_results*/results.csv + solutions/

Thank you!

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Questions?

`cvalverde@us.es · github.com/z72vamac/drone-cpp · articulo/main.tex`

Talk designed for 15–17 min. Slides in English, presentation in Spanish.